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CINEMATIC METADATA ORCHESTRATION AND INTELLIGENECE PLATFORM

INDIVIDUAL PROJECT: ASSIGNMENT 2

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Git Hub Repo link:

[https://github.com/Leo08012007/cinematic-metadata orchestration-platform](https://github.com/Leo08012007/cinematic-metadata-orchestration-platform)

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1. PROBLEM STATEMENT

With the rapid growth of OTT platforms and digital entertainment services, a huge volume of cinematic metadata is generated every day. This includes movie information, genres, cast details, ratings, reviews, popularity scores, and streaming statistics. Most of this data is scattered across multiple sources and is not organized in a structured manner for analytics and intelligent recommendations.

Traditional movie recommendation systems mainly focus on simple filtering techniques and often fail to provide meaningful insights, efficient data processing, and scalable analytics. In addition, many existing systems lack a modern cloud-based data engineering architecture capable of handling large datasets effectively.

To address these challenges, there is a need for a centralized analytics platform that can ingest, transform, store, analyze, and visualize cinematic metadata while providing intelligent movie recommendations using Microsoft Fabric.

1.1 Principal Difficulties

Data Scattered Across Multiple Sources

- Movie metadata is available from different datasets and APIs.
- Data exists in multiple formats and lacks proper integration.

Lack of Efficient Data Processing

- Raw datasets often contain missing values, duplicate records, and inconsistent formats.
- Manual data cleaning is time-consuming and error-prone.

Limited Analytical Capabilities

- Traditional systems provide basic movie search but lack advanced business analytics.
- Users cannot easily visualize trends such as genre popularity, rating distribution, or yearly movie releases.

Scalability Challenges

- Existing systems struggle to process continuously growing movie datasets.
- Large-scale analytical queries become slower as data volume increases.

Lack of Centralized Data Engineering Pipeline

- Data ingestion, transformation, storage, and reporting are often managed using separate tools.
- Maintaining multiple disconnected systems increases complexity.

Limited Recommendation Intelligence

- Many platforms provide only basic recommendations.
- They do not effectively utilize processed metadata to generate meaningful movie suggestions.

Need for Cloud-Based Analytics

- Organizations require scalable cloud platforms capable of handling data engineering workflows.
- Modern analytics solutions should support data warehousing, visualization, and business intelligence from a single platform.

2. PROPOSED SOLUTION

THE CINEMATIC METADATA ORCHESTRATION & INTELLIGENCE PLATFORM

The proposed solution is a cloud-based Movie Intelligence Platform developed using Microsoft Fabric and Azure technologies. The system is designed to ingest, transform, store, analyze, and visualize cinematic metadata using a modern data engineering architecture. It provides intelligent movie search, personalized recommendations, and interactive analytics while following the Medallion Architecture (Bronze, Silver, and Gold layers).

The platform integrates Microsoft Fabric Lakehouse, Warehouse, Power BI, and a web application to deliver an end-to-end solution for managing and analyzing movie datasets efficiently.

Microsoft Fabric Lakehouse

The Lakehouse acts as the centralized storage for all movie datasets.

Raw movie metadata is initially stored in the Bronze Layer, cleaned and standardized in the Silver Layer, and transformed into business-ready datasets in the Gold Layer for reporting and analytics.

Intelligent Movie Search and Recommendation

Users can search for movies based on title, genre, language, and release year.

The system provides personalized movie recommendations using processed metadata and similarity-based recommendation techniques.

Interactive Analytics Dashboard

Power BI dashboards provide visual insights into movie trends and analytics.

Users can explore genre distribution, rating analysis, language statistics, release trends, popularity metrics, and other business intelligence reports through interactive visualizations.

Modern and Responsive User Interface

The web application is developed using modern web technologies to provide an attractive and responsive interface.

Users can easily search movies, explore recommendations, and view analytics from desktop, tablet, and mobile devices.

Data Processing and Transformation

Microsoft Fabric Data Engineering capabilities are used to automate the complete ETL workflow. The system performs data ingestion, cleansing, transformation, validation, and loading using the Medallion Architecture to ensure high-quality analytical data.

Data Security and Reliability

The platform ensures secure storage and controlled access to movie metadata.

Role-based access control, cloud storage, and Microsoft Fabric security features help protect data while maintaining reliability and consistency.

Scalability and Future Enhancements

The proposed architecture is designed to support future enhancements without major modifications. Future versions of the system may include AI-powered recommendation models, real-time streaming analytics, user watchlists, sentiment analysis of movie reviews, multilingual search, and personalized user profiles.

3. THEME: CINEMATIC METADATA ORCHESTRATION & INTELLIGENCE PLATFORM

3.1 INTRODUCTION

3.1.1 Objective

The objective of this Software Requirements Specification (SRS) document is to identify and define the functional, non-functional, and design requirements of the **Cinematic Metadata Orchestration & Intelligence Platform**.

The project aims to develop a cloud-based movie analytics and recommendation platform using **Microsoft Fabric, Azure, Python, SQL, Power BI**, and modern web technologies. The system will implement the Medallion Architecture (Bronze, Silver, and Gold layers) to ingest, transform, analyze, and visualize cinematic metadata while providing an intelligent and user-friendly movie exploration experience.

3.1.2 Extent

The Cinematic Metadata Orchestration & Intelligence Platform enables users to search and explore movie information through a responsive web application while providing powerful analytical capabilities using Microsoft Fabric.

The system includes:

- Movie metadata ingestion from external datasets.
- Data transformation using Bronze, Silver, and Gold layers.
- Centralized storage using Microsoft Fabric Lakehouse.
- Analytical reporting through Power BI dashboards.
- Intelligent movie search and recommendation.

- REST API integration between the backend and frontend.
- Responsive web application for users to explore movie insights.

3.1.3 The Viewers

This Software Requirements Specification (SRS) document is intended for:

- Developers
- Data Engineers
- Project Managers
- UI/UX Designers
- Test Engineers
- Business Analysts
- Faculty Mentors
- Stakeholders involved in the project

Definitions, Acronyms, and Abbreviations

- **SRS** – Software Requirements Specification
- **UI** – User Interface
- **UX** – User Experience
- **API** – Application Programming Interface
- **ETL** – Extract, Transform, Load
- **BI** – Business Intelligence
- **SQL** – Structured Query Language
- **REST** – Representational State Transfer

- **DP-600** – Microsoft Certified: Fabric Analytics Engineer Associate
- **Lakehouse** – Unified architecture combining Data Lake and Data Warehouse
- **Power BI** – Microsoft's Business Intelligence and Data Visualization Platform

3.2 SYNOPSIS

3.2.1 A Look at the Product

The Cinematic Metadata Orchestration & Intelligence Platform is a cloud-based analytics system designed to process, analyze, and visualize large-scale movie metadata using Microsoft Fabric.

The platform combines modern Data Engineering, Business Intelligence, and Software Engineering concepts into a single integrated solution. Users can search movies, receive intelligent recommendations, and explore interactive dashboards generated from processed datasets.

3.2.2 Features of the Product

- Modern and responsive landing page.
- Intelligent movie search and filtering.
- Personalized movie recommendations.
- Interactive Power BI dashboards.
- Microsoft Fabric Lakehouse implementation.
- Bronze, Silver, and Gold data processing layers.
- REST API integration.
- Responsive design for desktop.

- Secure data management.
- Cloud-based analytics and reporting.

3.2.3 User Classes and Qualities

Regular Users

Users who search movies, explore recommendations, and view analytical insights through the web application.

Administrators

Users responsible for managing datasets, monitoring data pipelines, refreshing reports, and maintaining the analytics platform.

3.2.4 Environment of Operation

Frontend

- Next.js
- React
- Tailwind CSS
- TypeScript

Backend

- FastAPI
- Python

Database & Analytics

- Microsoft Fabric Lakehouse
- Azure SQL

Business Intelligence

- Microsoft Power BI

Cloud Platform

- Microsoft Azure
- Microsoft Fabric

3.2.5 Limitations in Design and Implementation

- The application should be lightweight and responsive.
- The system should provide fast search and analytical query performance.
- The platform should support all modern web browsers.
- The user interface should be mobile-friendly and responsive.
- The implementation should follow Microsoft Fabric best practices.
- The codebase should remain modular, maintainable, and well documented.
- The system currently uses a static movie dataset and does not process real-time streaming data.
- Advanced AI-based recommendation models are reserved for future enhancements.

4. FEATURES OF THE SYSTEM

4.1 Landing Page

- Modern, responsive, and visually appealing homepage.
- Attractive movie-themed hero section with featured movies.
- Easy navigation through menus and quick access buttons.
- Search bar for instant movie search.
- Featured movie exploration section.
- Optimized for fast loading and search engine visibility.

4.2 Movie Search Module

- Search movies using title, genre, language, release year, cast, or director.
- Advanced filtering options for accurate movie discovery.
- Quick search with dynamic suggestions.
- Displays detailed movie information including ratings, overview, runtime, and genres.
- Supports responsive search experience across all devices.

4.3 Movie Discovery Module

- Enables users to explore movies using metadata stored in the Microsoft Fabric Lakehouse and Azure SQL Database.

- Provides movie discovery through title search and filtering based on language, release year, ratings, and popularity.
- Displays structured movie metadata including genres, overview, runtime, release date, vote average, and vote count.
- Supports fast retrieval of movie information through REST APIs developed using FastAPI.
- Designed to support AI-powered recommendation models as a future enhancement.
-

4.4 Movie Details Module

- Complete movie information page.
- Displays movie poster, synopsis, genres, cast, crew, ratings, release date, runtime, and language.
- Shows popularity score and user ratings.
- Provides links to recommended similar movies.
- Clean and user-friendly movie detail interface.

4.5 User Dashboard

- Personalized dashboard for registered users.
- Displays recently viewed movies.

4.6 Analytics Dashboard (Power BI)

- Interactive Power BI dashboards.
- Genre distribution analysis.
- Movie rating analysis.

- Language-wise movie statistics.
- Release year trend analysis.
- Popularity and revenue visualization.
- Dynamic filters and drill-down reports.

4.7 User Interface Design (UI)

Include wireframes and UI mockups for:

- Landing Page
- Movie Search Page
- Movie Details Page
- Recommendation Page
- User Dashboard
- Power BI Analytics Dashboard
- Login and Registration Pages (if implemented)
- Responsive Mobile View

5.FUNCTIONAL REQUIREMENTS OF THE SYSTEM

Functional requirements describe the major functions and services that the Cinematic Metadata Orchestration & Intelligence Platform must provide. The system shall support efficient movie metadata processing, intelligent recommendations, analytics, and a user-friendly web interface.

5.1 Movie Search

- Users should be able to search movies using title, genre, language, release year, cast, or director.
- The system should display movie details such as title, overview, rating, release date, runtime, genres, and popularity.
- Search results should be displayed quickly with filtering and sorting options.

5.2 Movie Recommendation

- The system should recommend similar movies based on processed metadata.
- Recommendations should consider genre similarity, ratings, popularity, and other metadata attributes.

5.3 Analytics Dashboard

- The system should provide interactive Power BI dashboards.
- Users should be able to visualize movie trends, genre distribution, rating analysis, language statistics, and release year analysis.
- Dashboards should support filtering and interactive exploration of data.

- Reports should reflect the latest processed data available in Microsoft Fabric.

5.4 Data Processing Pipeline

- The system should ingest raw movie datasets into Microsoft Fabric.
- Raw data should be stored in the Bronze Layer.
- Cleaned and standardized data should be maintained in the Silver Layer.
- Business-ready analytical data should be generated in the Gold Layer.
- Data should be stored in the Fabric Lakehouse and Warehouse for reporting.

5.5 Responsive User Interface

- The application should provide a seamless experience on desktop, tablet, and mobile devices.
- The landing page should provide easy navigation to movie search, recommendations, and analytics.

5.6 Error Handling and Validation

- The system should validate user search inputs.
- Appropriate messages should be displayed when no matching movies are found.
- Invalid API requests should return meaningful error responses.
- Data validation should be performed before loading datasets into Microsoft Fabric.

5.7 Security Requirements

- The application should use secure communication protocols for data transmission.
- Access to Microsoft Fabric resources should be restricted to authorized users.
- User authentication mechanisms should protect sensitive application features.
- Data should be protected against unauthorized access through appropriate security controls.

6. NON-FUNCTIONAL REQUIREMENTS OF THE SYSTEM

Non-functional requirements define the quality attributes, performance expectations, security measures, scalability, and operational characteristics of the Cinematic Metadata Orchestration & Intelligence Platform.

6.1 Performance Requirements

- The application should load within 3 seconds under normal network conditions.
- Movie search results should be displayed with minimal response time.
- Microsoft Fabric data processing pipelines should execute efficiently for the provided datasets.
- Power BI dashboards should render quickly for interactive analysis.
- The system should support multiple concurrent users without significant performance degradation.

6.2 Scalability

- The platform should support increasing volumes of movie metadata.
- The architecture should allow additional datasets to be integrated without major redesign.
- Future versions should support real-time data ingestion and streaming analytics.
- The recommendation engine should be extendable with AI and Machine Learning models.

6.3 Reliability and Availability

- The platform should provide reliable access to movie data and analytical reports.
- Data processing pipelines should complete successfully without data loss.
- Microsoft Fabric resources should ensure consistent storage and retrieval of processed data.

6.6 Maintainability and Flexibility

- The project should follow a modular architecture.
- Source code should be clean, organized, and properly documented.
- The system should support future feature additions without significant architectural changes.
- Separate frontend, backend, and data engineering modules should simplify maintenance.

6.7 Portability and Compatibility

- The application should work correctly on major web browsers such as Chrome, Edge, Firefox, and Safari.
- The interface should adapt to desktops, tablets, and smartphone

7. MODULE DESCRIPTIONS

The Cinematic Metadata Orchestration & Intelligence Platform consists of multiple functional modules that collectively implement an end-to-end data engineering pipeline using Microsoft Fabric. Each module performs a specific role in data ingestion, processing, storage, analytics, and web application deployment.

7.1 Data Ingestion Module

Purpose:

- To ingest raw TMDB movie datasets into Microsoft Fabric Lakehouse.

Key Features:

- Import TMDB Movies and Credits datasets.
- Store raw CSV files in the Lakehouse.
- Preserve original dataset integrity.
- Prepare data for Bronze Layer processing.

7.2 Bronze Layer Module

Purpose:

- To store raw movie metadata in structured Lakehouse tables.

Key Features:

- Load raw datasets into managed Lakehouse tables.
- Preserve original data without modification.
- Validate successful data ingestion.
- Provide source tables for downstream processing.

7.3 Silver Layer Module

Purpose:

- To clean and standardize the Bronze Layer data.

Key Features:

- Handle missing values.
- Validate duplicate records.
- Standardize data types.
- Store cleaned data in the Silver Layer.

7.4 Gold Layer Module

Purpose:

- To prepare analytics-ready datasets.

Key Features:

- Generate refined analytical datasets.
- Create business metrics.
- Optimize data for reporting.
- Store processed data in the Gold Layer.

7.5 Movie Management Module

Purpose:

- To enable users to browse and explore movie metadata.

Key Features:

- Display movie collection.
- Search movies by title.

- Filter by language, rating, release year, and popularity.
- View detailed movie information.
- Retrieve movie data through FastAPI.

7.6 Analytics Dashboard Module

Purpose:

- To visualize movie analytics using Microsoft Power BI.

Key Features:

- Interactive Power BI dashboard.
- Revenue analysis.
- Budget analysis.
- Popularity insights.
- Genre analysis.
- Rating analysis.

7.7 Web Application Module

Purpose:

- To provide a user-friendly interface for movie exploration.

Key Features:

- Responsive Next.js frontend.
- Home page.
- Movies page.
- Dashboard page.
- Analytics page.

- About page.
- Navigation and filtering.

7.8 Backend API Module

Purpose:

- To provide REST APIs for communication between the frontend and Azure SQL Database.

Key Features:

- FastAPI REST APIs.
- Azure SQL connectivity.
- SQLAlchemy integration.
- JSON responses.
- Secure API communication.

7.9 Deployment Module

Purpose:

- To deploy the application using cloud services.

Key Features:

- Frontend deployment using Vercel.
- Backend deployment using Render.
- Azure SQL integration.
- Cloud-based application hosting.
- GitHub version control.

8. SYSTEM MODELS AND DIAGRAMS

The System Models and Diagrams section provides a visual representation of the Cinematic Metadata Orchestration & Intelligence Platform. These diagrams illustrate the system architecture, data flow, user interactions, and internal processing. They help developers, testers, and stakeholders understand the overall workflow and functionality of the system.

The following diagrams describe different aspects of the proposed system.

8.1 USE CASE DIAGRAM

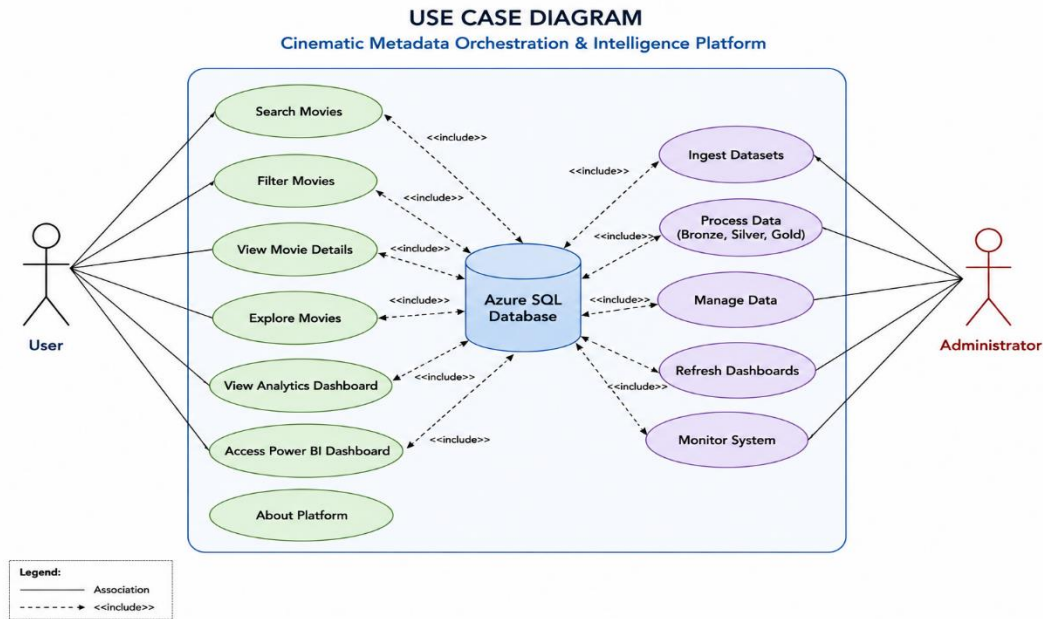
The Use Case Diagram illustrates the interaction between the users and the Cinematic Metadata Orchestration & Intelligence Platform. It provides a high-level view of the system functionalities available to users and administrators.

What it Shows

- User interactions with the web application.
- Administrator interactions for managing the platform.
- Major functionalities provided by the system.

Importance

- Provides a clear overview of system functionalities.
- Helps identify user requirements.
- Simplifies understanding of system operations.



8.2 FLOWCHART

The Flowchart illustrates the complete workflow of the system from data ingestion to website deployment.

What it Shows

- Data ingestion from TMDB datasets.
- Bronze, Silver, and Gold layer processing.
- Azure SQL Database integration.
- Power BI dashboard generation.
- Website deployment using Render and Vercel.

Importance

- Represents the end-to-end workflow.
- Simplifies system understanding.
- Helps identify processing stages.

FLOWCHART

Cinematic Metadata Orchestration & Intelligence Platform

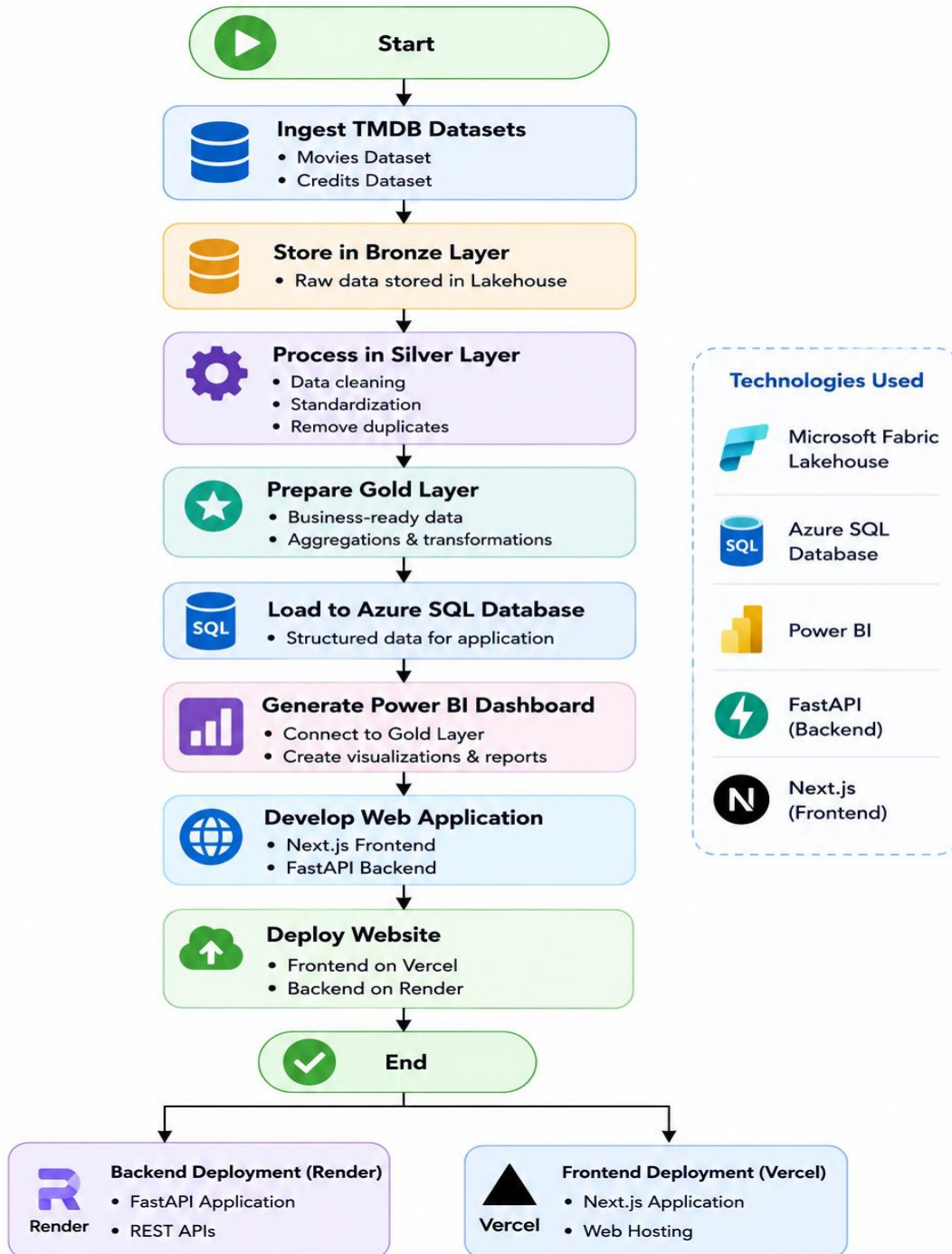


Fig 2: Flowchart

8.3 SEQUENCE DIAGRAM

The Sequence Diagram shows the interaction between the user, frontend, backend, Azure SQL Database, and Power BI dashboard.

What it Shows

- User requests.
- API communication.
- Database queries.
- Response flow.
- Dashboard access.

Importance

- Describes communication between system components.
- Shows the sequence of operations.
- Helps understand request processing.

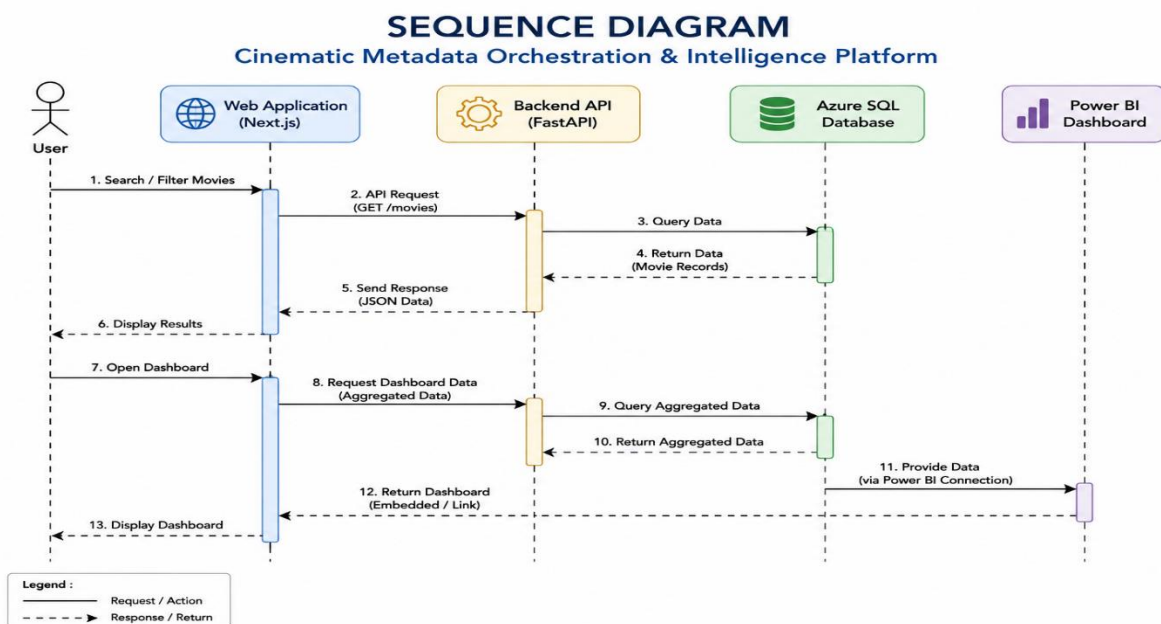


Fig 3: Sequence Diagram

8.4 STATE FLOW CHART

The State Flow Chart represents the different stages of movie data processing throughout the analytics pipeline.

What it Shows

- Raw data state.
- Data validation.
- Data transformation.
- Analytics-ready data.
- Website deployment.

Importance

- Illustrates the data lifecycle.
- Simplifies data processing flow.

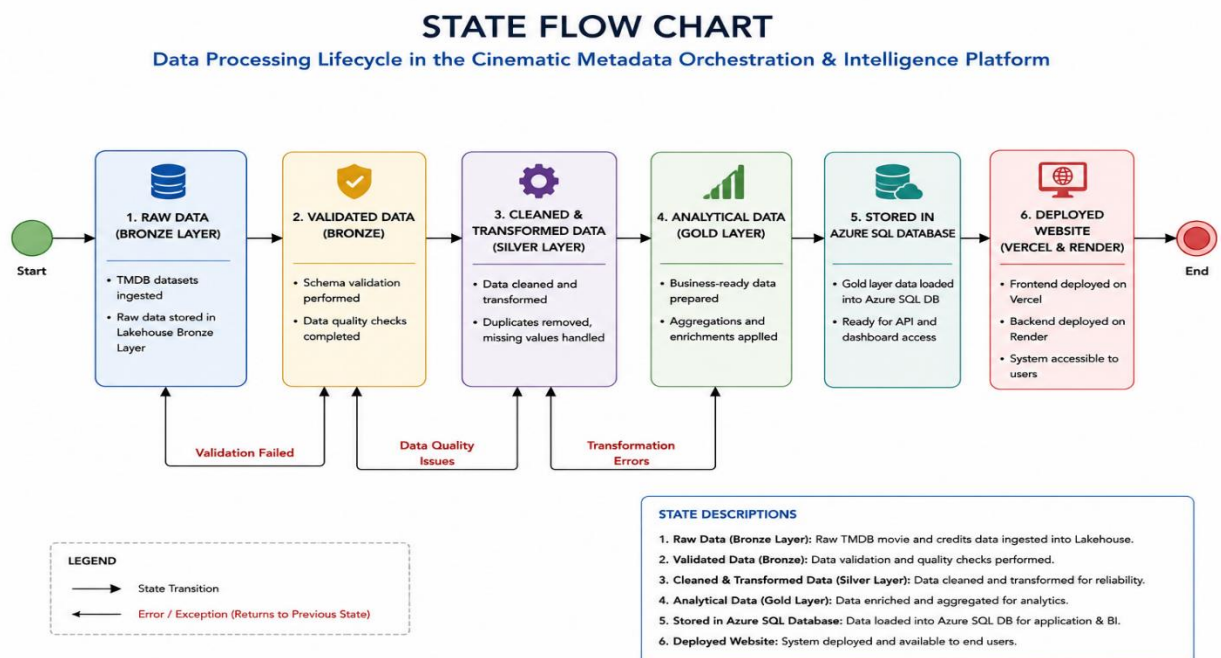


Fig 4: State Flow Chart

8.5 CLASS DIAGRAM

The Class Diagram represents the logical structure of the application by illustrating the main classes, their attributes, methods, and relationships. It provides an overview of how movie-related entities such as movies, genres, languages, and production companies are organized within the system.

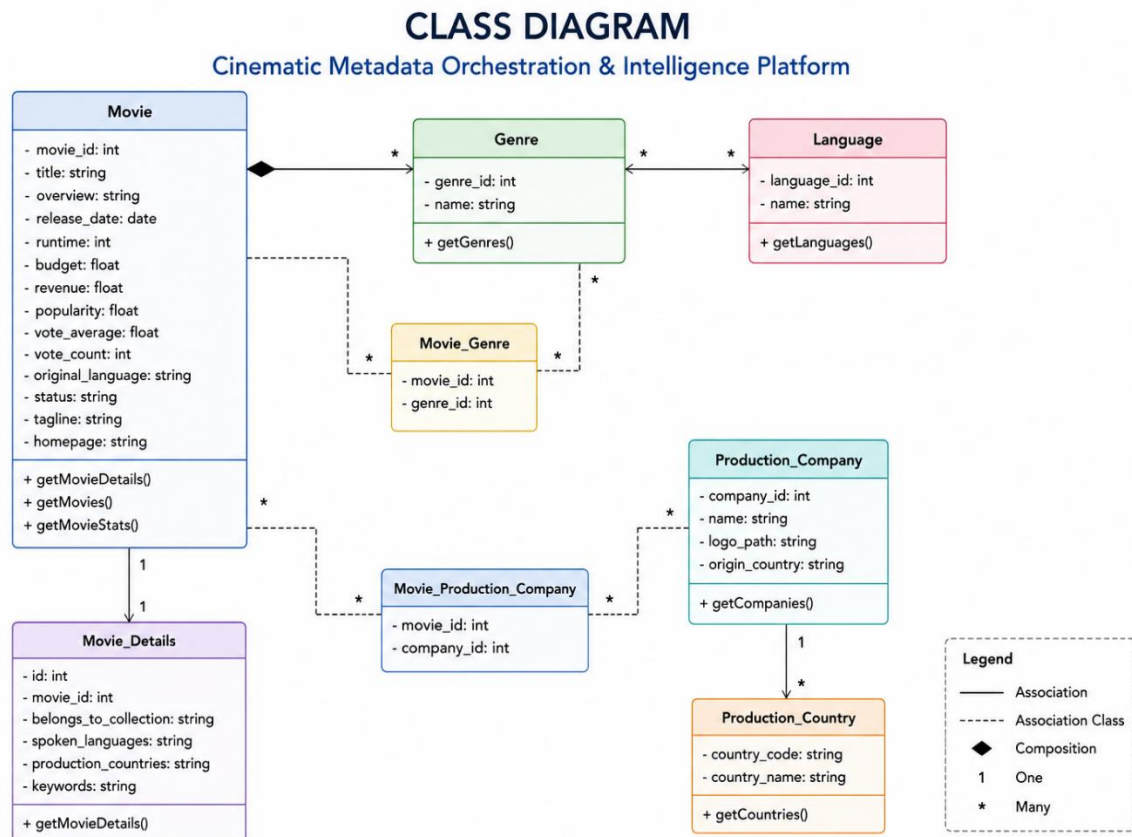


Fig 5: Class Diagram

9. CONCLUSION

In conclusion, this Software Requirements Specification (SRS) document presents a comprehensive overview of the **Cinematic Metadata Orchestration & Intelligence Platform**, defining its objectives, system architecture, functional requirements, non-functional requirements, and implementation strategy. It serves as a valuable reference for developers, testers, and stakeholders by providing a clear understanding of the project's scope, expected functionality, and overall system design.

The proposed system successfully integrates **Microsoft Fabric, Lakehouse architecture, Azure SQL Database, FastAPI, Next.js, and Power BI** to build a scalable cloud-based movie analytics platform. The implementation follows the Medallion Architecture by organizing data into Bronze, Silver, and Gold layers, enabling efficient data ingestion, transformation, and analytical processing. The web application provides users with an intuitive interface to explore movie metadata, search and filter movies, and visualize insights through interactive dashboards.

This document also defines the major system modules, diagrams, workflow, deployment architecture, and data processing pipeline, establishing a strong foundation for development, testing, deployment. Furthermore, the platform has been designed with scalability and extensibility in mind, allowing future enhancements such as AI-powered movie recommendations, advanced predictive analytics, real-time data integration, and additional visualization capabilities. Overall, this SRS provides a complete blueprint for developing a robust, efficient, and user-friendly cinematic analytics platform that meets both academic objectives and modern cloud-based data engineering practices

REFERENCES

Get started with Microsoft Fabric

<https://learn.microsoft.com/en-us/training/paths/get-started-fabric/>

- **Introduction to end-to-end analytics using Microsoft Fabric**
<https://learn.microsoft.com/en-us/training/modules/introduction-end-analytics-use-microsoft-fabric/>
- **Get started with lakehouses in Microsoft Fabric**
<https://learn.microsoft.com/en-us/training/modules/get-started-lakehouses/>
- **Use Apache Spark in Microsoft Fabric**
<https://learn.microsoft.com/en-us/training/modules/use-apache-spark-work-files-lakehouse/>
- **Work with Delta Lake tables in Microsoft Fabric**
<https://learn.microsoft.com/en-us/training/modules/work-delta-lake-tables-fabric/>
- **Orchestrate processes and data movement with Microsoft Fabric**
<https://learn.microsoft.com/en-us/training/modules/use-data-factory-pipelines-fabric/>
- **Ingest Data with Dataflows Gen2 in Microsoft Fabric**
<https://learn.microsoft.com/en-us/training/modules/use-dataflow-gen-2-fabric/>
- **Get started with data warehouses in Microsoft Fabric**
<https://learn.microsoft.com/en-us/training/modules/get-started-data-warehouse/>
- **Get started with Real-Time Intelligence in Microsoft Fabric**
<https://learn.microsoft.com/en-us/training/modules/get-started-kusto-fabric/>
- **Get started with data science in Microsoft Fabric**
<https://learn.microsoft.com/en-us/training/modules/get-started-data-science-fabric/>
- **Administer a Microsoft Fabric environment**
<https://learn.microsoft.com/en-us/training/modules/administer-fabric/>